

RatherKnow — Scoring Specification

Status: as-built, generated from the shipped code (not from documentation). **Date:** June 2026 · **Algo version stamped on every result:** `rk-1.0.0` **Source of truth for each section is named inline** so any claim here can be checked against the file.

This document exists to support a decision about changing the scoring presentation (stens vs percentiles vs something else). It describes, exactly, what each product does today: what is collected, how it is combined, what is published, and — critically — **what each instrument's evidence base can and cannot justify publishing.**

0. The constraints any change has to respect

Four rules are load-bearing. A presentation change that breaks one of them is a claims-register change, not a design change.

#	Rule	Where it bites
1	Scoring is frozen in v1. Parity/E2E tests gate every deploy; no math refactors.	Changing a <i>scale of presentation</i> is possible without touching scoring. Changing a <i>formula</i> is not, in v1.
2	Results are snapshotted with <code>algo_version</code> and never recomputed.	Old reports keep their old numbers. Any new scale must be derivable at read time, or versioned.
3	No banded scores before norms exist to justify them (refusal #3, "an unjustifiable number").	This is the whole percentile question. See §6.
4	"Validated" applies only to Personality and EI.	Presentation must not imply norm-referencing where there is none.

The short answer to the percentile question: percentiles are norm-referenced statements about a population. Exactly one instrument here has a population norm table — the Personality Mirror. See §6 for what each instrument could legitimately publish.

1. Personality Mirror — 130 items — *Established*

Files: `services/p150_lite.py`, `constants/p150_data.py`, `constants/p150_norms_snapshot.json`

Collection

- 130 Likert items, 1–5 (Strongly disagree → Strongly agree).
- 120 personality items: **15 factors × 8 items each**.
- 10 social-desirability validity items (do not enter any factor score).
- Reverse-keyed items are listed in `P150_REVERSED_ITEMS`; inversion is `6 - answer`.

Factor scoring (the actual measurement)

1. For each factor, sum its 8 items after reversal → **raw score, range 8–40**.
2. Convert raw → **sten 1–10** by table lookup in the frozen norms snapshot (`p150_norms_snapshot.json`, exported from the parent product's effective bands). Example, factor A (Warmth): raw 8–10 → sten 1; 11–12 → 2; 13–15 → 3; 16–18 → 4; 19–22 → 5; 23–25 → 6; 26–... → 7+. **Bands are per-factor, not shared.**
3. Sten label: 1–2 Very Low, 3–4 Low, 5–6 Average, 7–8 High, 9–10 Very High.
4. Missing answer defaults to 3 (the scale midpoint).

Global dimensions (derived, not measured)

Five second-order composites over the factor **stems**:

```
global = 5.5 + gain × Σ( weight × (factor_stem − 5.5) )    then clamped/rounded to 1–10
```

Dimension	Equation (factor: weight)	Gain
Extraversion	A +0.3, F +0.4, H +0.4, N −0.3, Q2 −0.5	1.0
Anxiety / Neuroticism	C −0.4, L +0.3, O +0.4, Q4 +0.4	1.0
Receptivity	A +0.3, I +0.5, M +0.3, Q1 +0.4	1.0
Independence	E +0.5, H +0.4, L +0.2, Q1 +0.3	1.0
Self-Control	F −0.3, G +0.4, M −0.3, Q3 +0.4	1.0

All gains are **1.0** (uncalibrated). GLOBAL_CALIBRATION._meta records why: "*n=1 gains trialled, not shipped (no population skew).*" Receptivity and Self-Control are flagged in-product as **known residuals** against other 16PF-style instruments. Receptivity is the same axis other reports publish as Tough-Mindedness, read from the opposite pole.

Validity indices (published, never hidden)

- **Social desirability**: count of the 10 checks answered in the flattering direction (≥ 4 , or ≤ 2 if reverse-keyed). NORMAL < 4 · ELEVATED 4–6 · HIGH ≥ 7 .
- **Central tendency**: % of all 130 answers at the midpoint. NORMAL < 40 · ELEVATED 40–54 · HIGH ≥ 55 .
- **Strengths** = factors with stem ≥ 8 . **Blind spots** = factors with stem ≤ 3 .

What the evidence supports

Calibrated per-factor norm bands exist → this is the **only** instrument here where a norm-referenced statement (percentile, population comparison) is defensible.

2. El Mirror — 140 items — *Established*

Files: routes/mirror_v2.py::_score_eq, constants/eimirror_data.py

Collection

140 Likert items, 1–5, across **4 domains** → **14 sub-dimensions** → **10 items each**. Roughly a third are reverse-keyed (EIMIRROR_REVERSE_ITEMS, inversion 6 − answer). All 140 are required; a missing answer defaults to 3.

Scoring

- **Sub-dimension score** = mean of its 10 reversed-corrected items → 1.00–5.00 (2 dp).
- **Domain score** = mean of all items in the domain (item-weighted, not a mean of sub-means).
- **Overall** = unweighted mean of the four domain scores.
- **Bands** (identical at all three levels): **High** ≥ 4.0 · **Moderate** ≥ 3.0 · **Developing** < 3.0 .
- Strengths = top 3 sub-dimensions; growth areas = bottom 3.

What the evidence supports

The bands are **published descriptive thresholds** — fixed cut-points on the raw 1–5 scale. There is **no norm table**, so nothing here is norm-referenced. A percentile would be an invented number.

3. Essential Mirror — 50 items × 2 lenses — *Developmental* (produces the Delta)

Files: `services/essential_scoring.py`, `routes/mirror_v2.py::_score_essential`,
`constants/essential_data.json`

Collection

The same 50 statements answered twice: once as **self**, once as the **ideal partner** (item ids are namespaced `self:1...50` and `ideal:1...50`). Likert 1–5. All 100 required.

Archetype scoring (per lens)

- 6 archetypes; each is keyed to **10 of the 50 items** (items are shared across archetypes).
- Answers map through a **non-linear point table**: 5→10, 4→7, 3→5, 2→2, 1→0 (reverse-keyed items invert first: 6 – answer).
- Archetype raw = sum over its 10 items (max 100); **percentage = raw / max × 100**.
- **Primary** = highest raw; **secondary** = second highest. The primary+secondary pair keys the narrative from `compatibility_matrix`.

Display dimensions (per lens)

Five fixed 10-item blocks — Values (1–10), Lifestyle (11–20), Conflict (21–30), Intellectual (31–40), Emotional (41–50) — each scored as $\text{sum} / (10 \times 5) \times 100$, rounded to a whole number. **Note the deliberate difference**: dimensions use the raw Likert sum; archetypes use the point table.

The Delta (the headline measurement)

```
delta[archetype] = ideal_percentage - self_percentage      (signed, 1 dp)
delta.overall    = mean( |delta[archetype]| ) over all 6   (1 dp)
delta.biggest    = archetype with the largest |delta|
dimension_gaps[d] = ideal_dimension[d] - self_dimension[d]
```

Sign convention: **positive** = you want more of it than you are. **Shadow** = the primary archetype's declared shadow counterpart, with its warning text.

What the evidence supports

Items and the archetype framework are ours. Early internal consistency only (archetype core: exploratory, $n=90$, α 0.68–0.74); no test–retest, **no population norms**. Percentages here are **proportions of a maximum possible score**, not percentiles — a distinction worth protecting in any redesign, because they look identical to a consumer.

4. Closeness Mirror (MI-AS-36) — 36 items + 1 check — *Developmental*

Files: `services/closeness_scoring.py`, `constants/closeness_bank_1_0_0.json`

Collection

36 items on a **7-point** scale in a **fixed presentation order** (part of the instrument spec — deliberately not randomised), plus one instructed-response validity item (`VAL-IR`). Keying is balanced: 9 forward / 9 reverse per dimension. Reverse inversion is 8 – answer.

Scoring

- Two dimensions of 18 items each: **anxiety** (items 1–18, published as *Reassurance*) and **avoidance** (items 19–36, published as *Closeness/distance*).
- Dimension score = **mean of answered items**, 1 dp. Prorated for up to 2 missing; **≥3 missing** → **status: not_scored** and no value is published.
- 12 facets of 3 items each (A1–A6, V1–V6), scored as a mean, and **suppressed unless ≥2 answered**.

Validity → confidence (nothing is deleted or silently corrected)

Check	Trigger
Instructed response	answer ≠ the bank's expected value
Inconsistency	Σ absolute difference over 3 keyed pairs ≥ 9
Long string	longest run of identical consecutive answers ≥ 7
Time floor	mean response time < 4000 ms

confidence = high (0 flags) · moderate (1) · low (≥ 2). Published on the result and in the PDF.

What the evidence supports

No norms exist, which is exactly why the result is a point on two continuous axes with **no bands, no quadrants, no types and no percentiles**. Pilot statistics ($n = 300$) are planned; when they exist, computed values will be published with sample size and date.

5. Flag Check — 9 questions — not an instrument

File: routes/mirror_v2.py: FLAG_ITEMS, complete_reflection

8 situation questions, each with 4 options mapping to a category: 1 → reprice · 2 → explain · 3 → late · 4 → none (didn't happen / can't recall), plus one safety question (f-safety: 1 no · 2 unsure · 3 yes).

```
informative = count(reprice) + count(explain) + count(late)
if informative < 3      → pattern = "insufficient"
else if top two tie    → pattern = "mixed" (+ the tied pair)
else                  → pattern = the modal category
```

Output contains no score, no band, no grade, no risk label — permanently. The safety answer selects a referral block; it never contributes to the pattern. Any change that gives this a number would turn a reflection into an audit of a relationship and inherit an evidence burden it cannot carry.

6. The percentile question, instrument by instrument

A percentile is a claim about a population: *"you scored higher than N% of people."* It requires a norm sample. Here is what each instrument can honestly support today.

Instrument	Norms?	Percentile defensible?	What it could publish instead
Personality Mirror	Yes — per-factor sten bands	Yes , for the 15 factors	Percentile band, or "higher than about N% of people", derived from sten
Personality globals	Derived from normed factors	Only as an approximation , and it inherits the Receptivity/Self-Control residual	Better shown as position + the equation behind it
El Mirror	No — fixed cut-points	No	The 1–5 score plus its published threshold (High/Moderate/Developing)

Essential Mirror	No	No	% of maximum (as now) and the Delta, clearly labelled as <i>not</i> percentiles
Closeness Mirror	No (n=300 pilot planned)	No	Position on the axis with the confidence label (as now)
Flag Check	n/a	Never	Nothing. It stays unscored.

If you want percentiles on the Personality Mirror

Stens are already normalised (mean 5.5, SD 2 by construction), so a sten → percentile mid-band mapping is arithmetic, requires no new norm data, and can be derived **at read time** — meaning snapshots stay immutable and old reports do not change:

Sten	1	2	3	4	5	6	7	8	9	10
Approx. percentile	2	5	12	27	40	60	73	88	95	98
% of people in band	2.3	4.4	9.2	15.0	19.1	19.1	15.0	9.2	4.4	2.3

Three cautions worth deciding on deliberately:

1. **Precision inflation.** A sten band is wide; "73rd percentile" sounds like a measurement to one person. "Higher than about 7 in 10 people" carries the same information without the false precision. Recommend banded/approximate language, never a decimal.
2. **Bidirectional factors.** Half these factors are not "more is better" (Vigilance, Privateness, Apprehension). A percentile invites a league-table reading of a trait that has costs at both poles. Any percentile presentation needs the two pole labels kept visibly attached.
3. **Comparability.** Dropping the sten entirely breaks comparison with every other 16PF-style report a user might hold — including the paired-sheet comparison that exposed the Self-Control residual. Recommend percentile as the headline, **sten retained as the underlying figure**.

What a scale change does *not* fix

Presentation is not measurement. Changing to percentiles will not move the Receptivity/Self-Control residual (that lives in the composite equations, §1), and will not create norms for the other three instruments. Those need a calibration sample — for the composites, $n \geq 200$ paired profiles; for Closeness, the planned $n = 300$ pilot.

7. Where a redesign is free, and where it is not

Free (presentation only, no scoring change, no register change):

- Percentile display for Personality factors, derived at read time from the existing sten.
- Any graphic treatment: bell curve with position marked, pole-to-pole bars, radar for EI domains, the existing two-axis dot plot for Closeness.
- Wording of labels, ordering, emphasis, print layout.

Not free (needs a deliberate decision, and in some cases a register update):

- Percentiles or bands on EI, Essential or Closeness → breaches refusal #3 while unnormed.
- Any change to a formula, weight, norm band, point table or threshold in §§1–4 → breaks hard rule 1 and the parity tests; needs a new `algo_version`, and old snapshots must keep their old numbers.
- Giving the Flag Check any number at all → breaks hard rule 5, permanently off the table.

Recommendation if you want the consumer upgrade now: percentile-led presentation on the Personality Mirror only (sten retained underneath, poles kept attached, approximate language), richer graphics across all four, and an explicit one-line note on the other three explaining why they show a position rather than a percentile. That is honest, ships without touching scoring, and is the strongest version of the "why can't I have a percentile here?" answer: because we don't have the norms yet, and we'd rather say so.

8. Verification gates protecting all of the above

Gate	What it locks
<code>tests/_e2e_test.py</code>	End-to-end scoring across all four instruments + Flag Check
<code>backend/tests/test_composites.py</code>	The five global equations, all gains at 1.0, a reference profile
<code>tests/test_locked_copy.py</code>	Locked claims copy (hash per string) + "validated" usage lint
<code>backend/tests/</code> (116 tests)	API behaviour, ownership, immutability, PDF content

Every stored result carries `algo_version: rk-1.0.0`. Results are written once (`$setOnInsert`) and never recomputed, so a future scoring change cannot rewrite history.